

Material Selection Analysis

Copper for Motor, Cables, and Power Supply Components



Reason for Material Choice:

High Electrical Conductivity (second only to silver):

Means less energy lost as heat, making systems more efficient.

Essential for low-voltage, high-speed digital signals (like in microcontrollers) and high-current power applications (like motors and PSUs).

High Thermal Conductivity:

Helps dissipate heat, critical in preventing component failure.

Ductility & Malleability:

Can be drawn into thin wires or etched into precise traces.

Ideal for flexible wiring, coils in motors, and tight PCB layouts.

Corrosion Resistance:

Ensures long life and stable performance over time, especially in open-air or humid environments.

Recyclability:

Copper is one of the most economically viable metals to recycle, which helps partially offset its high extraction impact.

Copper is essential for our hawk pulley system's electrical components due to its exceptional electrical conductivity, which ensures efficient power transmission with minimal energy losses. For the motor components, copper windings provide the electromagnetic field necessary for motor operation with excellent thermal conductivity to dissipate heat generated during operation. For the cables, copper offers superior current-carrying capacity in a minimal cross-section, helping keep our system lightweight while maintaining performance. The material's ductility allows for flexibility in your cabling, accommodating the movement inherent in a pulley system.

Extraction: Moderate energy demand at 60 MJ/kg and water usage at 330 L/kg. Copper mining generates substantial waste rock, tailings, and acid mine drainage. Ore grades are declining globally, increasing the environmental footprint per unit of copper.

Use: Highly durable with minimal degradation during normal use. Superior conductivity reduces energy losses in electrical systems. Copper wires can last from 70-100 years on average (Understanding the Lifespan of Electrical Components in Your Home, n.d.).

End-of-Life Analysis: When the hawk system reaches end-of-life, the copper components have excellent recovery potential. Under Singapore's Extended Producer Responsibility scheme for e-waste implemented in 2021, electrical components like motors and power supplies have designated collection channels (National Environment Agency, 2021). ALBA E-Waste Smart Recycling, Singapore's appointed PRO (Producer Responsibility Organization), operates collection points throughout the island (ALBA E-Waste Smart Recycling, 2023). Copper's high value (approximately 8-10 SGD per kilogram for scrap) creates strong economic

incentives for recovery. Singapore's e-waste recycling network can recover approximately 95% of copper from electronic components, representing significantly better environmental outcomes than landfilling.

Environmental Impact:

Mining pollution which is caused by habitat destruction, heavy metal leaching, water pollution while extracting copper ores. E-waste is also generated if not recycled properly after end-of-life. High energy use in smelting and processing where fossil fuels are still primarily used for these purposes.

PLA for 3D-Printed Chassis and Hawk Head



Reason for Material Choice:

Biodegradable & Bio-based

It is made from renewable resources like corn starch or sugarcane (unlike ABS or PETG which are fossil-based). Under industrial composting conditions, PLA can break down into water and CO₂.

Easy to Print

It has a low melting temperature of $\sim 180\text{--}220^{\circ}\text{C}$ which saves energy during printing

Good Dimensional Accuracy

It holds shape well which makes it suitable for developing mounts, brackets. Relatively rigid and hard, though brittle under impact. It is able to handle relative tugging from the rope and hawk.

PLA is used for our 3D-printed chassis and hawk head due to its excellent printability, allowing us to create complex geometric shapes with good dimensional stability. The material provides sufficient strength for non-load-bearing decorative elements while remaining relatively lightweight. PLA's ability to accept detailing makes it ideal for creating realistic features in the hawk head. Unlike ABS, PLA printing produces fewer potentially harmful emissions during the printing process, making it safer for home or workshop production, making it easier to be commercialised.

Production: With moderate energy requirements (47.2 MJ/kg) and lower water usage (21.8 L/kg), PLA has a smaller production footprint than petroleum-based alternatives. Its plant-based origin (typically corn starch) represents a renewable resource advantage.

Use: In Singapore's high humidity environment, PLA components in our pulley system will degrade very quickly, typically within a year. (Lucia, 2025) It requires protective coating to prevent gradual hygroscopic degradation. When properly sealed, PLA components should maintain integrity throughout the product's intended life. Thus, with proper maintenance we can extend the usage of our equipment.

End-of-Life Analysis: When our hawk system reaches end-of-life, the PLA components will most likely be processed through Singapore's waste-to-energy incineration system. Unlike in regions with industrial composting, Singapore lacks the infrastructure needed for proper PLA biodegradation. (Ghomi et al., 2021) Thus, in the end, it will be sent for incineration. Research at Nanyang Technological University has explored enhanced PLA recycling methods potentially applicable to future waste streams (Nanyang Technological University, 2021). For maximum sustainability, we are considering designing components for easy

disassembly to facilitate potential future recycling as infrastructure develops.

Environmental Impact:

Emissions during production (from growing and processing corn/starch) are still prevalent, where greenhouse gases are still released. Microplastics are also being released when degraded which can affect soil quality.(Effect of polylactic acid microplastics on soil properties, soil microbials and plant growth, 2023) Degradation of PLA requires industrial composting which is not truly biodegradable at home.

Mitigation:

Encourage the use of recycled PLA filament (for our case, in education institutions or in the general sense, industries). If we were to commercialise our product, we should continue with 3d printing instead of the conventional injection molding. It helps that parts we printed are modular and reusable, so that even when one component reaches its end-of-life we do not have to reprint the whole item, which helps to reduce PLA use and wastage.

Aluminum for Power Supply Casing, Power Supply Components, and Motor



Reason for Material Choice:

Excellent heat dissipation:

Aluminium helps to passively cool components which is important as power loss from the set-up is translated to heat. Being able to reduce overheating allows us to increase the lifespan of our product. It also helps to protect the sensitive electronics within safe temperatures.

Non-corrosive & durable:

Ideal for long-lasting, outdoor/industrial use. Especially in the case of our canteen, it is exposed to the environment. Due to the humidity and heat of Singapore, we want our products to be able to last longer in these environments to prevent it from reaching its end-of-life too soon.

Lightweight:

Easier to install our motors as it would not weigh down the whole set-up and in the event of it falling, it would help to reduce the risks.

Electromagnetic shielding:

It is important as our set-up is out in the open and it needs to be connected at all times to the laptop and webcam for it to be functional. The EM shielding helps to protect signal integrity and prevent disruption such as EM from other electronic devices.

Aluminum is ideal for our power supply casing and motor components due to its excellent strength-to-weight ratio, which helps minimize the overall weight of our ceiling-mounted system while maintaining structural integrity. The material's great thermal conductivity efficiently dissipates heat generated by the power supply and motor during operation, preventing overheating in Singapore's high temperature. Aluminum's corrosion resistance also ensures longevity even in Singapore's high-humidity environment, while its electromagnetic shielding properties help reduce electrical interference that could affect our system's performance. The material is also easily machined and formed, facilitating the manufacturing of complex component shapes.

Production: With high embodied energy (202 MJ/kg) and water usage (1,100 L/kg), aluminum production is resource-intensive. However, our

application leverages aluminum's functional advantages in relatively small quantities for specific components where its properties are most valuable.

Use: During operation, aluminum components help optimize our system's energy efficiency through effective heat dissipation, potentially extending component lifespan and reducing failure rates. The average lifespan of aluminium parts is 30-40 years which is relatively long, thus, we can maximise its usage by recycling before it reaches its end-of-life (Understanding the Lifespan of Electrical Components in Your Home, n.d.).

End-of-Life Analysis: When our hawk pulley system reaches end-of-life, the aluminum components have excellent recycling potential. Non-ferrous metals like aluminum achieve recycling rates of approximately 85-90% in Singapore according to NEA data (National Environment Agency, 2023). Singapore's regulated waste collection system facilitates separation of valuable metals (*Towards Zero Waste*, n.d.). Companies like Cimelia Resource Recovery operate specialized facilities for non-ferrous metal recovery. Recycling aluminum saves approximately 95% of the energy compared to primary production, representing significant environmental benefits. (International Aluminium Institute, 2024) Designing our system for easy disassembly to facilitate aluminum recovery at end-of-life.

Environmental Impact:

Mining bauxite (aluminium ore) can lead to deforestation and red mud waste. Water pollution occurs when mine tailings (toxic slurry left after extraction) often contain heavy metals like arsenic, mercury, and lead, which can contaminate local water supply.

If not recycled properly after use, disposal of metals adds to metal waste/e-waste. Moreover, Aluminium and copper mix in electronics can make material separation difficult, which makes recycling an even more challenging endeavor. Manufacturing involves precise machining, which can generate waste shavings. As such, there are several environmental risks at all stages of its life-cycle.

Mitigation:

We can use recycled aluminium, recycle casing post-use. We also need to take the initiative to separate our equipment properly before disposing it in the appropriate recycling venues. Also, due to the longevity of the material,

we can promote reusing of the aluminium (eg. making jewellery from old components) instead of going straight to recycling as although recycling is a lot better than incineration in terms of environmental impact, it still requires processing.

Styrofoam for Hawk Body



Reason for Material Choice:

Lightweight:

It is important to reduce load on the track to allow minimal drag or tension on the string. Also due to its low density, we can make the hawk large without compromising by making it heavy, which helps in saving

Flexible and safe:

It is also relatively safe and flexible, if it were to bump into someone, the likelihood of injury is very low. And it makes carving out the shape to be less physically intensive.

Cheap:

Styrofoam is very cheap to obtain and it makes it easier in our prototyping stages due to budget constraints.

Styrofoam provides extremely lightweight construction, which is critical for our mounted system where excess weight could create installation challenges or safety concerns. The material offers shape retention while minimizing the load on the mounting hardware and motor system. It can be easily carved and shaped to create realistic hawk features.

Production: With high embodied energy (85-105 MJ/kg) and moderate water usage (130-200 L/kg), styrofoam production is resource-intensive. However, since our application uses relatively small quantities for a decorative element, the total impact is moderated. Singapore imports all styrofoam, so production impacts occur outside its borders.

Use: The outdoor environment exposes it to UV degradation that would accelerate breakdown. It leads to photodegradation such as discoloration and yellowing (*Effects of UVC irradiation on polystyrene for healthcare packaging: Study by FTIR and Raman spectroscopy with thermoluminescence* 2024) as well as embrittlement and loss of strength (*Influence of ultraviolet radiation on polystyrene* 2021). This is detrimental if we were to continue using this in outdoor settings in the long term.

End-of-Life Analysis: When the hawk pulley system reaches end-of-life, the styrofoam body will likely enter Singapore's waste-to-energy stream. NEA data shows polystyrene recycling rates remain very low at around 4% (National Environment Agency, 2023). In Singapore's waste-to-energy plants, styrofoam can contribute to energy recovery, with its high carbon and hydrogen content yielding approximately 46 MJ per kg when incinerated (Federation, n.d.). The resulting ash will be transported to Semakau Landfill.

Environmental Impact:

Although styrofoam is cheap, its environmental cost is very high, especially at its end of life. Styrofoam does not biodegrade and can persist in the environment for over 500 years (Daugherty & Daugherty, 2017). In the US, around 75% of styrofoam end up in landfills (assuming percentage of polystyrene is similar to total plastic percentage) (*Plastics: Material-Specific Data | US EPA*, 2024). It breaks down into microplastics, which contaminate soil, waterways, and oceans, harming wildlife - styrofoam that end up in

waterways are mistaken for food by wildlife which clogs their digestive system and lead to their death (Daugherty & Daugherty, 2017).

Mitigation:

In future iterations, especially if we want to produce the hawks in larger quantities, we have to use more environmentally friendly alternatives. Green alternatives like mycelium or molded pulp packaging can be used instead. A study conducted in Germany reported that hemp-based mycelium composites have a carbon footprint of approximately 0.3668 kg CO₂ emitted per kg, which is significantly lower than that of polystyrene (Patel, D., & Joshi, G. , 2023). Other than that, Molded pulp packaging boasts an 82.6% recycling rate, is compostable, and has low carbon emissions, positioning it as a more eco-friendly alternative to polystyrene (THECA. ,2021).

PVC for Cable Insulators



Reason for Material Choice:

High dielectric strength and resistivity:

Good for both low and medium voltage applications. For PVC, its dielectric strength is typically around 40 kV/mm which means it can handle a decently high voltage before it wears out, this is very important to protect us from electric shocks. Its high resistivity helps mitigate short-circuit risks by requiring significantly elevated voltages to permit current leakage.

Heat and Flame Resistance:

Can handle temperatures up to 105°C in many formulations. Electrical resistance causes power loss that is translated to heat, PVC can help to protect us from the heat.

Self-extinguishing:

Contains chlorine, which helps slow flame spread during a fire - this is particularly important to tackle electrical fires.

Mechanical Durability:

Tough and abrasion-resistant. Flexible enough for wiring, yet protective against impact and wear. This is important as we want our cables and wires to last as long as possible to prevent frequent changing, which can be costly and environmentally damaging.

PVC is well-suited for our hawk pulley system's cable insulation due to its excellent electrical insulation properties, which safely contain current while preventing short circuits in your moving system. The material offers good flexibility that accommodates the dynamic movement inherent in a pulley mechanism while remaining durable enough to withstand repeated flexing. PVC's natural flame-retardant properties improve safety in electrical applications, particularly important for a ceiling-mounted installation. Its weather and moisture resistance protects the copper conductors from Singapore's high humidity, while its relatively low cost helps maintain overall project affordability without compromising performance.

Production & Moulding: With moderate energy demand (62.5 MJ/kg) and water usage (198 L/kg), PVC production has notable impacts, though these occur outside Singapore. Our system uses minimal amounts specifically for electrical safety applications. 6.39 MJ/kg of energy and 7.22 L/kg of water is used in the moulding process, which is less resource intensive than the extraction. This is so as polymerisation to crack hydrocarbons requires higher temperature at 500°C than melting and blending PVC at 160-200°C.

Use: During operation, PVC insulation remains stable, with minimal leaching of additives in indoor applications. Its electrical and thermal insulating properties contribute to system safety throughout the product's life cycle.

End-of-Life Analysis: When our hawk pulley system reaches end-of-life lifespan of around 10-30 years (*Latest trends and challenges in PVC and copper recovery technologies for end-of-life thin cables*, 2023), the PVC cable insulation will likely enter Singapore's waste-to-energy stream. The relatively small quantity used in our project minimizes disposal impact. Singapore's WTE plants are equipped with advanced emission control systems that manage the chlorine content in PVC during incineration (National Environment Agency., 2022). There are also limited recycling initiatives, for example, few companies like Tidy E-Way Plastic specialize in recycling various plastics, including PVC, transforming them into high-quality resins for manufacturing (*Plastic Recycling Services & Solutions Singapore* | Tidy Group, 2023).

Environmental Impact:

Water & Air Pollution. Chlorine production releases mercury into waters and this kills wildlife and causes Wastewater from PVC plants may contain heavy metals and organochlorines. PVC is rarely used in its pure form—it's mixed with plasticizers like phthalates – some linked to hormone disruption , or stabilizers (e.g., lead or cadmium compounds – toxic heavy metals). These cause harm to humans and wildlife when exposed when there is leakage from landfills. Incineration of PVC also releases Hydrogen Chloride and Dioxins which reacts with water to form acid rain (*ICPE Polyvinyl Chloride (PVC), Dioxins, Acid Rain*, n.d.), this corrodes buildings, pollutes water bodies killing aquatic life and also causes respiratory and cardiovascular diseases.

Mitigation:

To reduce the emission of harmful byproducts like dioxins, more incineration plants can install technologies like catalytic filters which can achieve destruction efficiencies of 97% to 99%, significantly reducing the release of these toxic compounds into the environment (Corporation, n.d.). Otherwise production companies can also embark on greener alternatives like bio-based polypropylene(PE). Reducing or eliminating the production and thus use of PVC in products altogether. For example, a petrochemical company Braskem has embarked on using sugarcane to produce "I'm Green™" bio-PE, which is fully recyclable and can reduce the need to rush to incineration after the end of life.

Assumptions:

1. We assumed that highest energy consumed to ensure we include the worst case scenarios.
2. We also assumed the H-bridge and ESP32's energy emission to be negligible compared to the other objects due to them being a small portion of our whole set-up.
3. We also assumed that our materials used are based on real-world conditions and real world renewable resource consumption is depicted from Grantapack, ceteris paribus.
4. For our LCA, we assumed that all renewable resources like energy and water encompass all that is consumed in the supply-chain, otherwise it would be an even longer analysis.
5. For our materials analyses, we chose the materials that occupy the highest percentage used in our project as they contribute most towards emissions when used.

Conclusion:

Overall, our material choices were guided by both performance needs and sustainability. While some materials like copper and aluminum offer durability and efficiency, they come with high environmental costs. Others like PLA and Styrofoam are lightweight and easy to work with, but raise concerns around biodegradability and end-of-life disposal. By designing with modularity, reusability, and local recycling options in mind, we've tried to strike a balance between practicality and environmental impact. Moving forward, exploring greener alternatives like bio-based plastics or eco-foams can help us improve future iterations.

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